

RQL: A Portable Intermediate Representation for Retrieval Plans in RAG Systems

Incremental research draft — not a camera-ready submission

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16 September 2026

Draft status. Structured for growth into a citable article/thesis. Claims are labelled [ESTABLISHED], [HYPOTHESIS], or [PROVISIONAL]. No fabricated experimental results. Not claimed ready for NeurIPS, VLDB, or arXiv acceptance.

Abstract

Retrieval-augmented generation (RAG) systems compose embedding search, lexical retrieval, metadata and ACL filters, hybrid fusion, reranking, and sometimes graph or multi-hop expansion. Today those stages are usually glued together in application code against vendor-specific vector-database APIs. This draft proposes **RQL (Retrieval Query Language)** not merely as “SQL for vectors”, but as a **portable intermediate representation for retrieval plans**: a small algebra over scored evidence, plus cost-based physical planning under recall, latency, and ACL constraints, compiling through adapters to heterogeneous backends.

We separate [ESTABLISHED] survey material from [HYPOTHESIS] original design and from [PROVISIONAL] literature pending multimodal re-read. Careful multimodal passes over the FANNS survey of Lin et al. [Lin et al., 2025] (VSP/VJP/SJP/SSP; A1–A17), ACORN [Patel et al., 2024], VBASE [Zhang et al., 2023], and Filtered-DiskANN [Gollapudi et al., 2023] ground filtered-ANN physical strategies (PRE/POST/SUBGRAPH/ITERATIVE/SPECIALIZED/PARTITION/ROUTER); Cormack et al. [Cormack et al., 2009] grounds Fuse_{rrf} ; Montague and Aslam [Montague and Aslam, 2002] ground $\text{Fuse}_{condorcet}$ (pairwise-majority sort); Bruch et al. [Bruch et al., 2023] ground Fuse_{linear} / CC-vs-RRF delineation; Chen et al. [Chen et al., 2022] ground the zero-shot RRF-vs-linear-interp critique that resolves the Bruch disagreement as a setup conflict; ColBERT [Khattab and Zaharia, 2020] grounds MaxSim/late-interaction scoring for Search_{late} ; PLAID [Santhanam et al., 2022] grounds centroid-interaction / multi-stage pruning for late-interaction engines; MU-VERA [Dhulipala et al., 2024] grounds an $\text{FDE} \rightarrow \text{MIPS} + \text{MaxSim}$ -rerank physical rewrite (journals 0011–0014, 0016–0019). A docs-only vendor API matrix (journal 0020) [ESTABLISHED] the adapter surfaces RQL must compile to. We define an evaluation *plan* and reproducibility layout; we do **not** report unreproduced benchmark scores.

1 Introduction

1.1 Problem

Modern RAG stacks must retrieve *evidence*—chunks, passages, entities, paths—under mixed modalities: dense vectors, sparse/BM25 channels, structured predicates (including ACLs), and increasingly late-interaction or graph expansion [Lewis et al., 2020]. Vector databases expose powerful but **non-portable** interfaces (JSON filters, GraphQL operators, SQL extensions, Redis commands, Cypher search). As a result, accuracy, latency, and reliability become properties of brittle Python glue rather than of inspectable plans.

1.2 Motivation (accuracy, latency, reliability)

Accuracy. Hybrid fusion (e.g. reciprocal rank fusion [Cormack et al., 2009]), over-fetch+rerank, and multi-query rewriting improve recall only when stages are explicit and tunable.

Latency. Filter placement (pre vs. post vs. subgraph vs. specialised label graphs vs. iterator ANN) and ANN parameters dominate end-to-end cost [Patel et al., 2024, Gollapudi et al., 2023, Zhang et al., 2023].

Reliability. Silent empty results, client-side ACL filtering, and irreproducible efSearch folklore are operational failures; plans with EXPLAIN and mandatory predicate pushdown targets are the remedy [HYPOTHESIS].

1.3 Thesis statement

RQL is a portable IR for retrieval plans—an algebra over scored multisets of evidence together with a Cascades/Calcite-style physical planner that chooses filtered-ANN and fusion strategies under explicit recall/latency/ACL contracts, then compiles via polystore-like shims to concrete engines. The textual DSL is a frontend, not the contribution’s centre of mass [HYPOTHESIS].

1.4 Contributions (draft)

1. A clear problem formulation for portable retrieval plans (Section 4).
2. A hypothesized retrieval algebra and physical operator set (Sections 5–6) [HYPOTHESIS].
3. Compilation/adaptor doctrine inspired by Substrait and polystores (Section 7) [HYPOTHESIS].
4. An evaluation and reproducibility *plan* without fabricated metrics (Section 9).
5. Careful related-work positioning, with ACORN, VBASE, and Filtered-DiskANN multimodally re-read (Section 3).

2 Background

2.1 Approximate nearest neighbour search

Graph indices such as HNSW [Malkov and Yashunin, 2018] and disk-oriented graphs (DiskANN line) underpin production vector search. Search quality/latency is controlled by parameters (e.g. candidate list size), yielding recall–throughput curves rather than exact SQL cardinalities [ESTABLISHED].

2.2 Hybrid vector + predicate search

Applications need ANN jointly with structured predicates. Common strategies include **pre-filtering**, **post-filtering**, specialised low-cardinality equality indices, and **predicate-subgraph** traversal as in ACORN [Patel et al., 2024] [ESTABLISHED] (see Section 3).

2.3 Hybrid lexical–dense fusion

Reciprocal Rank Fusion (RRF) combines ranked lists without score calibration [Cormack et al., 2009] [ESTABLISHED] (journal 0011). Condorcet-fuse sorts candidates by pairwise majority over input rankings [Montague and Aslam, 2002] [ESTABLISHED] as the majoritarian rank-only sibling (journal 0018); Cormack AUTHORS later prefer RRF over Condorcet on their suites [PROVISIONAL] as unreproduced comparison. Bruch et al. [Bruch et al., 2023] analyse the sibling *convex combination* of normalised lexical and semantic scores $f_{\text{Convex}} = \alpha \phi_{\text{Sem}}(f_{\text{Sem}}) + (1 - \alpha) \phi_{\text{Lex}}(f_{\text{Lex}})$ (especially TM2C2 with theoretical

min-max ϕ_{tmm}) versus RRF, and argue RRF is parameter-sensitive while CC is sample-efficient to tune [ESTABLISHED] as literature mechanisms (journal 0016); author NDCG/Recall tables are unreproduced here [PROVISIONAL]. Chen et al. [Chen et al., 2022] prefer RRF over tuned min-max linear interpolation in a zero-shot hybrid setting (Recall@1K; journal 0017), disagreeing with Bruch’s TM2C2>RRF(60) NDCG claim — an [ESTABLISHED] setup-level literature conflict (see §6.4). Planner choice among Fuse_{rrf} , $\text{Fuse}_{\text{condorcet}}$, $\text{Fuse}_{\text{linear}}$, and Fuse_{ltr} remains workload-dependent [HYPOTHESIS].

2.4 Late interaction

ColBERT [Khattab and Zaharia, 2020] defines *late interaction*: independent encoders produce bags of contextualized embeddings E_q, E_d , scored by a sum of maximum similarities (MaxSim)

$$S_{q,d} = \sum_{i \in [|E_q|]} \max_{j \in [|E_d|]} E_{q_i} \cdot E_{d_j}^\top$$

(Eq. (3) in Khattab and Zaharia [2020]; cosine via L2-normalized dots; squared L2 also evaluated) [ESTABLISHED] (multimodal journal 0012). Document bags are offline-indexable; MaxSim is pruning-friendly for end-to-end retrieval. Subsequent engines add physical rewrites: PLAID’s centroid interaction and multi-stage centroid pruning funnel (candidate generation $\rightarrow$ pruned centroid MaxSim $\rightarrow$ unpruned centroid MaxSim $\rightarrow$ residual MaxSim) [Santhanam et al., 2022] [ESTABLISHED] as literature mechanisms (journal 0014; author latency tables unreproduced), and MUVERA Fixed Dimensional Encodings that reduce Chamfer/MaxSim search to single-vector MIPS plus MaxSim rerank [Dhulipala et al., 2024] [ESTABLISHED] as a literature mechanism (journal 0013); RQL compile packaging of the ColBERT $\rightarrow$ PLAID $\rightarrow$ MUVERA ladder remains [HYPOTHESIS].

2.5 RAG as a staged pipeline

A typical RAG retrieve path is: embed $\rightarrow$ search channels $\rightarrow$ filter $\rightarrow$ fuse $\rightarrow$ (rerank) $\rightarrow$ expand/traverse $\rightarrow$ generate. RQL aims to name these stages as plan operators rather than ad-hoc functions [HYPOTHESIS].

3 Related work

3.1 Vector database interfaces

Commercial and OSS systems expose vendor-specific query surfaces (metadata filters, hybrid search APIs, SQL extensions such as pgvector). A **docs-only adapter-surface survey** (2026-09-16; journal 0020; docs/09-vendor-api-matrix.md) tabulates filter DSLs, filter+ANN composition (PRE/POST/iterative/UNKNOWN), hybrid/BM25+dense, RRF and weighted fusion primitives, multi-vector/late interaction, query surfaces, and EXPLAIN/plan visibility for Qdrant, Elasticsearch/OpenSearch, Weaviate, Milvus, and pgvector (optional Pinecone/Redis) [ESTABLISHED] as a *documentation survey*—not a live bake-off. None of these vendors offers a cross-vendor ANSI-SQL analogue for multi-stage RAG plans [ESTABLISHED] (landscape notes + matrix). The hypothesis that RQL LogicalPlan compiles through negotiated shims onto these surfaces is stated in §7.3.

3.2 Filtered ANN / FANNS

Lin et al. survey (multimodal re-read). Lin et al. [Lin et al., 2025] formalise hybrid datasets/queries and propose a pruning-focused taxonomy—**VSP, VJP, SJP, SSP**—classifying 17 algorithm families (A1–A17; Figs. 1–2). They further argue that hybrid query difficulty depends on *selectivity and distribution* (ID/POD/OOD; Figs. 3–6), and highlight combining multiple FANNS algorithms with dynamic selection as a system-level direction (§6.3). Pass 1–5 notes (journal 0019; knowledge/reads/fanns-lin2025-2505.0650)

treat the **taxonomy and A1–A17 map** as [ESTABLISHED] survey structure grounding FilterExec packaging; author Fig. 3 recall curves remain unreproduced [PROVISIONAL].

ACORN (multimodal re-read). Patel et al. [Patel et al., 2024] argue that pre- and post-filtering are brittle under selectivity and query–predicate correlation, and that specialised equality indices constrain predicate cardinality. They propose denser, predicate-agnostic HNSW variants (ACORN- γ , ACORN-1) that *search the predicate subgraph* to approximate an oracle HNSW over surviving vectors. Our Pass 1–5 notes (journal 0006; OKF bundle under knowledge/reads/acorn-2403.04871/) treat author QPS claims as **not reproduced here** [ESTABLISHED] for the problem taxonomy; [PROVISIONAL] for unreproduced numeric speedups.

Filtered-DiskANN (multimodal re-read). Gollapudi et al. [Gollapudi et al., 2023] change *graph construction* (FilteredVamana / StitchedVamana) so edges respect geometry and labels, then search with label-aware greedy expansion. Pass 1–5 notes (journal 0010; knowledge/reads/filtered-diskann-www23/) ground FilterExec = SPECIALIZED for equality/label predicates; complex SQL-like filters remain an author-stated open problem [ESTABLISHED] for the specialised-graph idea; [PROVISIONAL] for unreproduced QPS curves.

VBASE (multimodal re-read). Zhang et al. [Zhang et al., 2023] formalise *relaxed monotonicity* and expose ANN traversal as Volcano Open/Next/Close inside PostgreSQL, enabling filter-during-traversal, multi-column TopK, range filters, and vector joins without a predicted over-fetch K' . Pass 1–5 notes (journal 0009; knowledge/reads/vbase-osdi23/) ground FilterExec = ITERATIVE and a concrete path for VSimJoin [ESTABLISHED] for the iterator architecture; [PROVISIONAL] for unreproduced latency claims.

3.3 Rank fusion (RRF and linear/CC)

Cormack et al. (multimodal re-read). Cormack, Clarke, and Büttcher [Cormack et al., 2009] define Reciprocal Rank Fusion as $RRFscore(d) = \sum_{r \in R} 1/(k + r(d))$ with $k = 60$ fixed after a pilot study. RRF is unsupervised and *rank-only* (no requirement that channel scores be calibrated), unlike CombMNZ. Pass 1–5 notes (journal 0011; knowledge/reads/rrf-cormack-sigir09/) treat the **formula and default k** as [ESTABLISHED] IR prior art for logical Fuse_{rrf}; author TREC/LETOR MAP tables are **not reproduced here** [PROVISIONAL] as unreproduced empirics.

Montague and Aslam [Montague and Aslam, 2002] introduce *Condorcet-fuse*: a majoritarian (Social Choice) fusion that sorts the candidate pool using a pairwise majority runoff over input rankings as the comparison function ($O(nk \log n)$; ranks only). Their Fig. 1 taxonomy places Condorcet-fuse beside Borda-fuse/rCombMNZ in the ranks-only / no-training cell, opposite CombMNZ on the score axis; weighted Condorcet and dependence filtering (correlated-run prune) are optional variants. Pass 1–5 notes (journal 0018; knowledge/reads/montague-aslam-cikm02-condorcet/) treat the **pairwise-majority sort mechanism and Fig. 1 fusion-input taxonomy** as [ESTABLISHED] literature for logical Fuse_{condorcet} as a Fuse-family sibling of Fuse_{rrf}; author TREC MAP / sign-test tables remain [PROVISIONAL] / AUTHOR-only. Cormack et al. later report RRF matching or beating Condorcet Fuse on their suites [ESTABLISHED] as an AUTHOR comparison (journal 0011) — motivating RRF as the portable default while retaining Condorcet as an optional majoritarian sibling [HYPOTHESIS].

Bruch et al. (multimodal re-read). Bruch, Gai, and Ingber [Bruch et al., 2023] analyse hybrid lexical–semantic fusion via a *convex combination* of normalised channel scores (TM2C2: theoretical min–max + α) versus RRF (including parametric η_{Lex} , η_{Sem} and RRF-CC). They argue CC learning is largely agnostic among monotone normalisers, RRF is parameter-sensitive, and CC is sample-efficient; on their suite they report TM2C2 beating default RRF on NDCG (disagreement with Chen et al.). Pass 1–5 notes (journal

0016; knowledge/reads/bruch-arxiv-2210.11934/) treat the **CC/TM2C2 formulas, normalisation role, and RRF-vs-CC delineation** as [ESTABLISHED] literature mechanisms for logical Fuse_{linear} (with Fuse_{ltr} as the learned multi-parameter end of the continuum); author Recall/NDCG tables remain [PROVISIONAL] / AUTHOR-only.

Chen et al. (cite-chase from Bruch [5]; multimodal re-read). Chen, Zhang, Lu, Bendersky, and Najork [Chen et al., 2022] study *zero-shot* hybrid retrieval and fuse lexical (+expansion) with deep (NPR) channels via RRF ($k=60$). They argue score linear interpolation needs min-max normalisation and α tuning that conflicts with zero-shot transfer; on Robust04 and TREC-COVID, their best-tuned min-max BM25+NPR interpolation still underperforms RRF(BM25, NPR) by about 3% relative Recall@1K (Fig. 2; primary metric Recall@1K/MAP, not NDCG). Pass 1–5 notes (journal 0017; knowledge/reads/chen-ecir2022-2201.10582/) treat the **RRF-for-zero-shot mechanism and the linear-interp critique** as [ESTABLISHED] literature; author Recall/MAP tables remain [PROVISIONAL] / AUTHOR-only.

Resolving Chen $\leftrightarrow$ Bruch (setup disagreement). The author-level disagreement is [ESTABLISHED] as a *literature conflict of setups*, not a single false claim: different metrics (Recall@1K vs NDCG), different linear forms (simple min-max α -interp vs TM2C2 ϕ_{tmm}), different objectives (zero-shot / no labels vs sample-efficient α with labels), and different channel/dataset suites. RQL packaging and the planner preference rule among $\{\text{Fuse}_{rrf}, \text{Fuse}_{linear}, \text{Fuse}_{ltr}\}$ remain [HYPOTHESIS] (see §6.4).

3.4 Late interaction / multi-vector retrieval

ColBERT (multimodal re-read). Khattab and Zaharia [Khattab and Zaharia, 2020] introduce contextualized late interaction: bags of BERT embeddings scored by MaxSim-sum (Eq. (3)). Pass 1–5 notes (journal 0012; knowledge/reads/colbert-sigir20/) treat the **MaxSim scoring definition and offline document-bag pattern** as [ESTABLISHED]; author MS MARCO MRR/latency tables are not reproduced here [PROVISIONAL].

PLAID (multimodal re-read). Santhanam et al. [Santhanam et al., 2022] accelerate ColBERTv2 late-interaction search with a four-stage funnel: centroid-based candidate generation, *centroid interaction* (approximate MaxSim over bags of centroid IDs using $S_{c,q} = CQ^\top$), *centroid pruning* by threshold t_{cs} , then residual decompression and exact MaxSim. Pass 1–5 notes (journal 0014; knowledge/reads/plaid-cikm22/) treat the **centroid-interaction / pruning pipeline (Fig. 5; Eqs. (2)–(5))** as [ESTABLISHED] literature mechanisms; author GPU/CPU speedup and MRR tables are not reproduced here [PROVISIONAL].

MUVERA (multimodal re-read). Dhulipala et al. [Dhulipala et al., 2024] reduce Chamfer/MaxSim multi-vector search to single-vector MIPS via Fixed Dimensional Encodings, then Chamfer rerank—contrasting PLAID’s multi-stage residual-preserving centroid pipeline [Santhanam et al., 2022]. Pass 1–5 notes (journal 0013; knowledge/reads/muvera-2405.19504/) treat the **FDE reduction + MIPS + MaxSim rerank mechanism** as [ESTABLISHED] literature; RQL rewrite packaging and unreproduced BEIR/PLAID latency numbers remain [HYPOTHESIS] / [PROVISIONAL].

3.5 IR query languages

Classical IR languages (Indri, Galago, CQL/SRU) treat ranking as an operator algebra [met, 2004] [PROVISIONAL] (URL-verified references; deep re-read pending). RQL steals composability, not surface syntax fidelity [HYPOTHESIS].

3.6 Optimisers, portable plans, polystores

Apache Calcite [Begoli et al., 2018] and Substrait [sub, 2024] motivate logical/physical separation and portable plan IR. BigDAWG-style polystores [Duggan et al., 2015] motivate islands + shims [PROVISIONAL]. BlinkDB [Agarwal et al., 2013] motivates explicit approximate budgets [PROVISIONAL].

3.7 Knowledge packaging (methodology, not runtime)

Google’s Open Knowledge Format (OKF) packages knowledge as markdown concepts with YAML frontmatter and graph-shaped links [okf, 2025]. We use OKF-*shaped* bundles for research notes. We do **not** claim OKF is part of the RQL execution runtime [ESTABLISHED].

3.8 What RQL is not

RQL is not “another vendor filter DSL”, not a guarantee of identical recall across HNSW libraries, and not a claim that SQL alone solves multi-stage RAG [HYPOTHESIS].

4 Problem formulation

4.1 Setting

Let collections store entities with payloads, optional graph links, and one or more retrieval channels (dense vectors, lexical fields, multi-vectors). A **retrieval plan** consumes a user/agent query (text and/or vectors), mandatory security predicates, and optional budgets, and returns a ranked evidence multiset.

4.2 Requirements

1. **Portability:** one logical plan maps to multiple backends via adapters.
2. **Filter correctness:** ACL/tenant predicates are hard constraints (fail closed).
3. **Explicit approximation:** recall/latency targets are first-class, not hidden knobs.
4. **Composability:** fusion, rewrite, expand, traverse, and similarity joins are plan nodes.
5. **Observability:** logical/physical EXPLAIN with provenance for evaluation harnesses.

4.3 Non-goals (this draft)

Identical numeric recall across ANN implementations; full DDL standardisation; replacing embedding model APIs; unreproduced SOTA claims.

5 RQL algebra (original sketch)

All operators in this section are [HYPOTHESIS] unless marked otherwise.

5.1 Typed evidence relation

An **evidence relation** E is a multiset of tuples with at least:

$$E \subseteq Id \times Payload \times Score \times Channel \\ \times Provenance \times Attr^*.$$

Informal signature: each tuple carries a channel-local score until a fuse operator. Mandatory ACL attributes (tenant, clearance, ...) are part of $Attr$ and must remain enforceable in every physical plan [HYPOTHESIS].

Channels. $Channel \in \{\text{dense, bm25, late, graph, fused, } \dots\}$. Scores are comparable *within* a channel; cross-channel comparison requires Fuse.

5.2 Logical operators (kernel)

Operator	Signature (sketch)	Notes
$\text{Search}_{\text{dense}}(q, k)$	$\rightarrow E$	ANN leaf
$\text{Search}_{\text{bm25}}(q, k)$	$\rightarrow E$	Lexical leaf
$\text{Search}_{\text{late}}(q, k)$	$\rightarrow E$	Late interaction / MaxSim-sum scoring [ESTABLISHED] [Khatab and Zaharia, 2020] (journal 0012); RQL op name / compile [HYPOTHESIS]
$\text{Filter}(P)$	$E \rightarrow E$	Predicates / ACL
Union	$E \times E \rightarrow E$	Multi-query
$\text{Fuse}_{\text{rrf}}(k_{\text{rrf}})$	$E^* \rightarrow E$	Formula $\sum 1/(k + r(d))$, default $k=60$ [ESTABLISHED] [Cormack et al., 2009] (journal 0011); RQL op name / compile [HYPOTHESIS]
$\text{Fuse}_{\text{condorcet}}$	$E^* \rightarrow E$	Pairwise-majority sort (Condorcet-fuse) [ESTABLISHED] as mechanism [Montague and Aslam, 2002] (journal 0018); RQL op / compile [HYPOTHESIS]
$\text{Fuse}_{\text{linear}}(\alpha)$	$E^* \rightarrow E$	Convex combo of normalised channel scores [ESTABLISHED] as mechanism [Bruch et al., 2023] (journal 0016); RQL op name / ϕ / compile [HYPOTHESIS]
$\text{Fuse}_{\text{ltr}}(\text{model})$	$E^* \rightarrow E$	Learned / multi-feature fuse beyond single α [HYPOTHESIS] (continuum with linear; not a Bruch TM2C2 claim)
$\text{Diversify}_{\text{mmr}}$	$E \rightarrow E$	Diversity
Rerank_m	$E \rightarrow E$	Cross-encoder etc.
Expand	$E \rightarrow E$	Parent/window
Rewrite	$q \rightarrow q^*$	HyDE / multi-query
Traverse_h	$E \rightarrow E$	Bounded graph hop
VSimJoin_θ	$E \times E \rightarrow E$	Similarity join

5.3 Rewrite examples (hypothesis, literature-grounded)

- **Post-filter brittleness (ACORN seed).** $\text{Filter}(P) \circ \text{Search}_{\text{dense}}(q, k)$ may be rewritten toward a physical SUBGRAPH or over-fetched POST plan when estimated selectivity / query–predicate correlation is adverse [Patel et al., 2024] [HYPOTHESIS] (choice is physical; logical Filter remains).
- **Iterator pushdown (VBASE seed).** When the adapter exposes Volcano-style ANN iteration, $\text{Filter}(P) \circ \text{Search}_{\text{dense}}$ may compile to $\text{FilterExec} = \text{ITERATIVE}$ (filter during Next) rather than TopK-then-drop [Zhang et al., 2023] [HYPOTHESIS].
- **Label-specialized graphs (Filtered-DiskANN seed).** Equality/label predicates with a capable filtered graph may compile to SPECIALIZED [Gollapudi et al., 2023] [HYPOTHESIS]; workload-stable partitions may compile to PARTITION and multi-algorithm backends to ROUTER [Lin et al., 2025] [HYPOTHESIS]; unsupported predicate shapes must not silently use specialised paths.

- **Hybrid fuse (RRF / linear / LTR).** Parallel Search_{dense} and Search_{bm25} (or other channels) remain separate until a fuse op. $\text{Fuse}_{rrf}(k_{rrf})$ with default $k_{rrf} = 60$ [Cormack et al., 2009] is the portable *rank-only* default when scores are incomparable [ESTABLISHED] (journal 0011). $\text{Fuse}_{condorcet}$ is the majoritarian rank-only sibling (pairwise-majority sort) [Montague and Aslam, 2002] [ESTABLISHED] as mechanism (journal 0018); offer when pairwise Condorcet semantics are desired, without displacing RRF as default [HYPOTHESIS] (Cormack AUTHORS prefer RRF on their suites). $\text{Fuse}_{linear}(\alpha)$ follows Bruch-style convex combination of normalised scores [Bruch et al., 2023] [ESTABLISHED] as mechanism (journal 0016; unreproduced NDCG [PROVISIONAL]); escalate to Fuse_{ltr} when a learned multi-feature model is justified [HYPOTHESIS]. Choosing among $\{rrf, condorcet, linear, ltr\}$ is physical/policy [HYPOTHESIS] (§6.4; Chen $\leftrightarrow$ Bruch setup disagreement [ESTABLISHED], journals 0016–0017).
- **Late-interaction leaf + rewrites (ColBERT / PLAID / MUVERA).** Logical Search_{late} carries MaxSim-sum semantics [Khattab and Zaharia, 2020] [ESTABLISHED]. Physical compile targets [HYPOTHESIS]: (i) native multi-vector MaxSim; (ii) PLAID-class centroid interaction + multi-stage prune then residual MaxSim [Santhanam et al., 2022] [ESTABLISHED] as the literature pipeline (journal 0014; unreproduced speedups remain [PROVISIONAL]); (iii) MUVERA FDE $\rightarrow$ MIPS then MaxSim/Chamfer rerank [Dhulipala et al., 2024] when only single-vector ANN is available (journal 0013). Missing multi-vector, PLAID, and FDE capabilities $\rightarrow$ refuse or EXPLAIN, not silent dense cosine substitution.
- **Similarity join.** $\text{VSimJoin}_{\theta}(E_1, E_2)$ may compile to nested Search_{dense} / range-iterator joins when ITERATIVE (or equivalent) exists [Zhang et al., 2023] [HYPOTHESIS]; otherwise to blocked nested loops with explicit cost — not claimed implemented.

5.4 Semantics sketch

Channels remain separate until fuse. Mandatory predicates in $P_{acl} \subseteq P$ must be enforced by physical plans that cannot strip them client-side [HYPOTHESIS]. Rewrite operators carry explicit token/latency cost attributes for the optimiser.

5.5 Delineation from survey material

RRF [Cormack et al., 2009] (formula + $k=60$; journal 0011), Condorcet-fuse [Montague and Aslam, 2002] (journal 0018), Bruch CC/TM2C2 linear fusion [Bruch et al., 2023] (journal 0016), Chen zero-shot RRF-vs-interp [Chen et al., 2022] (journal 0017), ColBERT MaxSim/late interaction [Khattab and Zaharia, 2020] (journal 0012), PLAID centroid interaction/pruning [Santhanam et al., 2022] (journal 0014), MUVERA FDE reduction [Dhulipala et al., 2024] (journal 0013), HNSW [Malkov and Yashunin, 2018], ACORN’s predicate-subgraph idea [Patel et al., 2024], VBASE’s relaxed-monotonicity iterators [Zhang et al., 2023], and Filtered-DiskANN’s label-aware graphs [Gollapudi et al., 2023] are prior art (multimodal notes: journals 0006, 0009–0014, 0016–0018). **Original to this draft:** packaging them as a single portable retrieval IR with explicit physical filter modes and budget contracts aimed at RAG reliability [HYPOTHESIS].

6 Physical planning (original sketch)

This section is [HYPOTHESIS] unless a citation marks established prior art.

6.1 Physical filter modes

Grounded in the FANNS survey taxonomy of Lin et al. [Lin et al., 2025] (pruning strategies VSP/VJP/SJP/SSP; families A1–A17; journal 0019) and multimodal re-reads of ACORN [Patel et al., 2024], Filtered-

DiskANN [Gollapudi et al., 2023], and VBASE [Zhang et al., 2023]:

FilterExec $\in$ {PRE, POST, ITERATIVE, SUBGRAPH, SPECIALIZED, PARTITION, ROUTER, AUTO}.

Survey taxonomy $\rightarrow$ modes (packaging). Lin et al. argue that coarse pre-/post-/in-filtering is too coarse and propose a pruning-focused classification [ESTABLISHED] as *survey structure* (Figs. 1–2):

- PRE $\leftarrow$ SSP / Pre-Filtering family (A12): exact scalar filter then scan the survivor set [Lin et al., 2025].
- POST $\leftarrow$ VSP / Post-Filtering family (A1): ANN top- K' then predicate; K' estimation is brittle under unknown selectivity [Lin et al., 2025, Patel et al., 2024].
- ITERATIVE $\leftarrow$ VBase (A2) as a VSP refinement with filter-aware updates and relaxed-monotonicity stopping [Lin et al., 2025, Zhang et al., 2023]; Volcano Open/Next/Close when the backend advertises iterators.
- SUBGRAPH $\leftarrow$ VJP / ACORN-class predicate-subgraph traversal (A4; AIRSHIP A3 related) [Lin et al., 2025, Patel et al., 2024].
- SPECIALIZED $\leftarrow$ VJP equality/range graph families (Filtered-DiskANN A9; SeRF/iRangeGraph A10–A11 class) when the predicate matches advertised capabilities [Lin et al., 2025, Gollapudi et al., 2023].
- PARTITION $\leftarrow$ SJP workload/filter partitions with per-subset FANNS (Milvus-Partition A13; HQI A14; related multi-subset SJP) [Lin et al., 2025].
- ROUTER $\leftarrow$ system-level *combination of multiple FANNS algorithms* with per-query selection (survey open direction §6.3; selectivity cost-model switches in ADBV/Milvus/VBase; ACORN $\rightarrow$ PRE when highly selective) [Lin et al., 2025] [ESTABLISHED] as a *literature direction*; RQL opcode packaging [HYPOTHESIS].
- AUTO $\leftarrow$ choose among *advertised* modes using stats (selectivity, specificity, query–predicate correlation / ID–POD–OOD distribution factor [Lin et al., 2025], empty-result risk); **never** violate ACL hard constraints [HYPOTHESIS].

Label discipline: VSP/VJP/SJP/SSP and A1–A17 are [ESTABLISHED] survey vocabulary; FilterExec names and the planner policy that binds them are [HYPOTHESIS]. Author recall curves (e.g. survey Fig. 3) are **not reproduced here**.

Capability negotiation. Adapters must advertise which modes exist. Missing SUBGRAPH/SPECIALIZED/PARTITION forces a documented fallback — the algebra must not pretend universal support [HYPOTHESIS]. EXPLAIN should surface FilterExec, survey pruning_strategy when known, selectivity estimates, and optional distribution factor.

6.2 Cost vector

Unlike classical selectivity-only costing, plans expose a vector:

1. Recall proxy (candidate depth, filter empty-result risk, low-specificity risk [Gollapudi et al., 2023], distribution-mismatch risk [Lin et al., 2025]),
2. Latency (p50/p95 targets),
3. ACL safety (boolean hard constraint),
4. Token/dollar budget (embeddings, rerankers, rewrites).

BlinkDB-style explicit budgets [Agarwal et al., 2013] motivate surface options such as RECALL TARGET / LATENCY [HYPOTHESIS].

6.3 Fusion and rewrite rules

Examples [HYPOTHESIS]:

- Independent channel searches $\rightarrow$ parallel schedule; choose fuse family per §6.4 [HYPOTHESIS] (policy grounded in Cormack/Montague/Chen/Bruch mechanisms [Cormack et al., 2009, Montague and Aslam, 2002, Chen et al., 2022, Bruch et al., 2023]).
- Search_{late} compile ladder: native MaxSim multi-vector [Khattab and Zaharia, 2020] [ESTABLISHED] scoring; else PLAID-class centroid interaction + multi-stage prune [Santhanam et al., 2022] [ESTABLISHED] as mechanism (journal 0014) if late_plaid advertised; else MUVERA FDE $\rightarrow$ MIPS + MAXSIM_RERANK [Dhulipala et al., 2024] if fde_mips capability exists (journal 0013); else fail closed with EXPLAIN (do not silently substitute single-vector cosine).
- Low selectivity + POST $\rightarrow$ rewrite toward ITERATIVE/SUBGRAPH/SPECIALIZED/PARTITION/over-fetch depending on capabilities and predicate shape [Lin et al., 2025, Patel et al., 2024].
- High selectivity + cheap survivors $\rightarrow$ prefer PRE (SSP) or ROUTER policies that switch to Pre-Filtering [Lin et al., 2025].
- VSimJoin $\rightarrow$ prefer iterator nested loops when ITERATIVE exists [Zhang et al., 2023].

6.4 When Fuse_{rrf} vs Fuse_{linear} (planner preference)

Chen et al. [Chen et al., 2022] and Bruch et al. [Bruch et al., 2023] disagree on which fuse wins empirically; the disagreement is [ESTABLISHED] as a literature conflict of *setups* (journal 0017 cite-chase): Recall@1K zero-shot RRF vs NDCG TM2C2, different linear forms, and different label budgets. We therefore treat the following as a planner [HYPOTHESIS] (not a reproduced metric claim):

- Prefer Fuse_{rrf}($k=60$) when channel scores are incomparable/uncalibrated, or when no (or vanishing) labeled queries exist to tune α/ϕ — Chen-style zero-shot portability [Cormack et al., 2009, Chen et al., 2022] [ESTABLISHED] as mechanism.
- Offer Fuse_{condorcet} (pairwise-majority sort) as a rank-only Fuse-family sibling when majoritarian semantics are desired [Montague and Aslam, 2002] [ESTABLISHED] as mechanism (journal 0018); do not displace RRF as the portable default (Cormack AUTHORS prefer RRF on their suites) [HYPOTHESIS].
- Prefer Fuse_{linear}(α) with documented ϕ (TM2C2-class theoretical min-max when available) when scores are comparable and a small labeled set can tune α — Bruch-style [Bruch et al., 2023] [ESTABLISHED] as mechanism.
- Escalate to Fuse_{ltr} when query features or multi-channel weights justify learning beyond a single α [HYPOTHESIS].
- EXPLAIN must surface fuse family, k or α/ϕ or Condorcet comparator, and input channels.

7 Compilation and adapters

[HYPOTHESIS] doctrine; Substrait/polystore inspiration is [PROVISIONAL]/[ESTABLISHED] as prior systems. Vendor API cells below are [ESTABLISHED] from official docs (2026-09-16 survey); the claim that RQL compiles cleanly through them is [HYPOTHESIS].

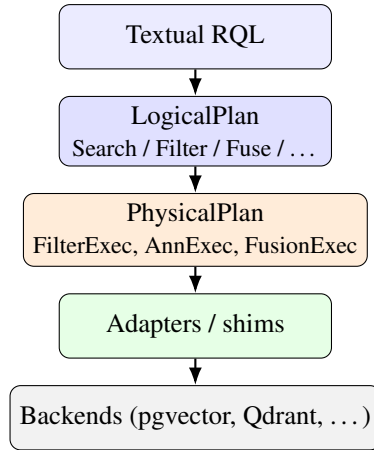


Figure 1: Hypothesized RQL compilation stack: text $\rightarrow$ logical plan $\rightarrow$ physical plan $\rightarrow$ adapters.

7.1 Layers

1. Parse textual RQL to `LogicalPlan`.
2. Optimise to `PhysicalPlan` with capability negotiation.
3. Emit backend calls through adapters (pgvector SQL, Qdrant Query API, Elasticsearch/OpenSearch DSL, Weaviate GraphQL, Milvus `hybrid_search`, ...).

7.2 Established adapter surface survey (docs-only, Sep 2026)

A docs-only survey of public vendor APIs (journal 0020; companion package `docs/09-vendor-api-matrix.md`; OKF knowledge/`reads/vendor-api-matrix-2026-09/`) records, without live cluster calls:

- **Filter DSLs** differ (Qdrant `JSON Filter`; ES/OS Query DSL; Weaviate `where`; Milvus `expr`; pgvector SQL `WHERE`).
- **Filter+ANN composition** is not uniform: Weaviate documents pre-filtering; Elasticsearch documents `knn.filter` during approximate kNN; pgvector documents post-filter on approx indexes plus iterative scans ($\geq 0.8.0$); Milvus API text states `expr` before ANN; Qdrant emphasises leaf filter propagation; Redis documents hybrid `BATCHES/ADHOC_BF` modes.
- **RRF** is first-class in Qdrant (`rrf`), Elasticsearch (`retriever.rrf`), OpenSearch (score-ranker RRF), and Milvus (`RRFRanker`); Weaviate exposes `rankedFusion`; pgvector/Pinecone often require client/SQL shims.
- **Weighted / score fusion** appears as Milvus `WeightedRanker`, Weaviate `relativeScoreFusion+alpha`, Qdrant `dbsf/formula`, ES `boosts/weighted RRF`—semantics are *not* interchangeable.
- **Multi-vector / late interaction** is documented for Qdrant multivector `rescore` and Weaviate multi-vector (ColBERT-style); elsewhere often **UNKNOWN** or nested dense only.
- **EXPLAIN / plan visibility**: PostgreSQL `EXPLAIN` (pgvector) and ES `Profile` / OpenSearch hybrid score explanation are clearly documented; several others remain **UNKNOWN**, so RQL `EXPLAIN` will often be adapter-synthesised.

URLs and UNKNOWN cells are cited in the package matrix; this section does not invent undocumented APIs [ESTABLISHED] as a *surface survey*.

7.3 Hypothesis: RQL compiles to these shims

We [HYPOTHESIS] that a single RQL LogicalPlan can target the above surfaces via capability negotiation: map $\text{Filter} \circ \text{Search}_{dense}$ to the backend’s PRE / POST+iterative / leaf-propagated / hybrid-batch mode; map Fuse_{rrf} to native RRF where present else an explicit client shim (ACL-marked); map Fuse_{linear} only when score-normalisation contracts match; map Search_{late} to native multivector paths or to the ColBERT→PLAID→MUVERA rewrite ladder. Live smoke verification remains a human P1 task—not claimed done here.

7.4 Capability negotiation

Profiles declare required features (hybrid+RRF, graph hop, late interaction, filter mode). Missing capabilities yield either a safe rewrite, a shim CAST (e.g. client-side fuse—explicitly marked expensive/unsafe for ACL), or a hard error [HYPOTHESIS].

7.5 Substrait adjacency

Substrait [sub, 2024] motivates a binary/portable plan encoding across engines. RQL may eventually emit a retrieval-oriented dialect or embed custom extensions; this draft does not claim a finished Substrait specification [HYPOTHESIS].

8 Applications to RAG patterns

Patterns below map to example programs in the companion repo (`examples/*.rql`).

- Dense + ACL metadata filter (reliability-first pushdown).
- **Hybrid BM25+dense with RRF / linear:** $\text{Fuse}_{rrf}(k=60)$ over channel-local rankings [Cormack et al., 2009] [ESTABLISHED] as the portable default when scores are incomparable; optional cross-encoder Rerank afterward (`examples/02-hybrid-rrf.rql`). Prefer $\text{Fuse}_{linear}(\alpha)$ (Bruch-style normalised convex combination [Bruch et al., 2023] [ESTABLISHED] as mechanism, journal 0016) when calibrated scores + small labels exist; keep RRF under zero-shot / uncalibrated scores [Chen et al., 2022] [ESTABLISHED] as the competing literature recommendation (journal 0017; preference rule [HYPOTHESIS], §6.4); optional $\text{Fuse}_{condorcet}$ (pairwise-majority sort) [Montague and Aslam, 2002] [ESTABLISHED] as mechanism (journal 0018) for majoritarian A/B without replacing RRF; escalate to Fuse_{ltr} for multi-feature learned fuse [HYPOTHESIS] (`examples/03-hybrid-linear-weights.rql`, `12-fuse-learned-explain.rql`).
- Multi-query / HyDE-style rewrite as plan nodes (costed) [PROVISIONAL].
- Parent expansion and bounded multi-hop graph retrieval [PROVISIONAL].
- **Late-interaction search:** logical Search_{late} with MaxSim-sum semantics [Khattab and Zaharia, 2020] [ESTABLISHED]; optional planner rewrite toward PLAID-class centroid interaction/pruning [Santhanam et al., 2022] [ESTABLISHED] as mechanism (journal 0014) or MUVERA FDE→MIPS+MaxSim rerank [Dhulipala et al., 2024] when the adapter lacks native multi-vector support [HYPOTHESIS] (`examples/09-colbert-late-interaction.rql`).

RQL’s value is making these patterns *comparable* under EXPLAIN and shared eval harnesses [HYPOTHESIS] — not inventing RRF or MaxSim themselves [ESTABLISHED].

9 Evaluation plan (no fabricated results)

This section specifies what we will measure, not results we have not run.

9.1 Datasets (planned)

- Retrieval: BEIR/MS MARCO-style collections as available under license; synthetic metadata predicates for filter selectivity sweeps.
- Filtered-ANN microbenchmarks aligned with published FANNS setups (e.g. SIFT-scale vectors with attribute predicates), following protocols from prior work such as ACORN [Patel et al., 2024]—**re-run**, do not copy tables.

HUMAN_TODO: dataset download, licenses, and digests — see `experiments/HUMAN_TODO.md` and `experiments/protocols/02-datasets.md`.

9.2 Metrics

Recall@ k , nDCG@ k , latency p50/p95, throughput (QPS) where applicable, **plan stability** (same RQL → same logical plan hash), and ACL violation count (must be zero).

9.3 Baselines

1. Hand-written SDK glue (LangChain/LlamaIndex-style scripts).
2. Vendor filter+hybrid APIs directly.
3. pgvector SQL hybrid patterns.
4. Fixed `FILTER_MODE` ablations vs AUTO (once implemented).

9.4 Ablation matrix (planned)

Channels $\in$ {dense, bm25, both}; fuse $\in$ {RRF, linear}; filter mode $\in$ {PRE, POST, AUTO}; rerank on/off; rewrite on/off.

9.5 What exists today (agent-run)

- Planned-run printer: `experiments/harness/print_planned_runs.py` (emits score=NOT_RUN only).
- Deterministic RRF unit test on fixed lists: `experiments/harness/test_rrf_fusion.py` → `experiments/results/rrf_unit_test.txt`.
- Deterministic weighted linear/CC fusion on fixed toy scores: `experiments/harness/test_linear_fusion.py` → `experiments/results/linear_fusion/unit_test.txt` (not an IR benchmark).

No end-to-end ANN, RAG, or human-judgment metrics are claimed.

9.6 What requires a human (Vijay)

GPU/CPU FANNS microbench, production/docker vector DB smokes, paid APIs, large datasets, and relevance judgments are listed as **HUMAN_TODO** items with exact protocols and expected in-repo artifact paths in `experiments/HUMAN_TODO.md`. Those results should be pushed to this same GitHub repository when available.

10 Threats to validity

- **Literature coverage:** only ACORN has completed multimodal Pass 1–5 in-repo; other citations remain [PROVISIONAL].
- **ANN nondeterminism:** recall is implementation-dependent; portable IR cannot claim bit-identical results.
- **Adapter impedance:** not all backends expose subgraph or hybrid fusion; shims may hide costs.
- **Optimism bias:** algebra elegance does not guarantee planner quality without measured cost models.

11 Conclusion

We framed RQL as a **portable retrieval-plan IR** for RAG—algebra plus physical planning under recall/latency/ACL—rather than as cosmetic SQL for vectors [HYPOTHESIS]. ACORN, VBASE, and Filtered-DiskANN multimodal re-reads [ESTABLISHED] filter strategy as a first-class physical decision [Patel et al., 2024, Zhang et al., 2023, Gollapudi et al., 2023]. Cormack RRF, Montague–Aslam Condorcet-fuse, Bruch CC/TM2C2, and Chen’s zero-shot RRF-vs-linear critique delineate the Fuse family (Fuse_{rrf} , $\text{Fuse}_{condorcet}$, Fuse_{linear}) [Cormack et al., 2009, Montague and Aslam, 2002, Bruch et al., 2023, Chen et al., 2022] [ESTABLISHED] as mechanisms (journals 0011, 0016–0018); the planner preference rule is [HYPOTHESIS] (§6.4). ColBERT→PLAID→MUVERA grounds the late-interaction ladder [Khattab and Zaharia, 2020, Santhanam et al., 2022, Dhulipala et al., 2024]. The immediate path is: freeze a LogicalPlan schema against the Sep 2026 vendor matrix (journal 0020; docs-only [ESTABLISHED] adapter-surface survey), implement two adapters, and execute the evaluation plan with published artifacts—without shortcutting integrity.

A Reproducibility checklist

1. Scripts and configs under `experiments/` with pinned dependency files when benchmarks are added.
2. Dataset license notes and download digests (**HUMAN_TODO:** `experiments/protocols/02-datasets.r`).
3. Seeded RNGs where stochasticity exists; report ANN library versions in `ENV.txt`.
4. Store logical/physical plan JSON beside metrics (no orphaned scores).
5. Separate AUTHOR_CLAIM citations from REPRODUCED runs in result tables.
6. Research notes: journal + OKF-shaped bundles for paper reads.
7. Hand-off list for compute/credentials beyond the agent sandbox: `experiments/HUMAN_TODOS.md`.

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